

```
%pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12
```

```
import numpy as np
from sklearn.linear_model import LinearRegression
```

```
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
Y = np.array([35, 45, 50, 52, 65, 72, 86, 89, 90, 96])
```

```
model = LinearRegression()
model.fit(X,Y)
```

▼ LinearRegression ⓘ ?
LinearRegression()

```
Prediction = model.predict([[5.5]])
print("Predicted value for 5.5 hours study is:", Prediction[0])
```

```
Predicted value for 5.5 hours study is: 68.0
```

```
import numpy as np
from sklearn.linear_model import LogisticRegression
```

```
x_animals = np.array([[4,5], [5,6], [20,15], [25,18], [3,4], [30,20]])
y_labels = np.array([0, 0, 1, 1, 0, 1])
```

```
classifier = LogisticRegression()
classifier.fit(x_animals, y_labels)
```

▼ LogisticRegression ⓘ ?
LogisticRegression()

```
new_animal = [[22, 12]]
result = classifier.predict(new_animal)
```

```
if result[0] == 1:  
    print("The animal is classified as a dog.")  
else:  
    print("The animal is classified as a cat.")
```

The animal is classified as a dog.

```
new_animal = [[15, 4]]  
result = classifier.predict(new_animal)  
if result[0] == 1:  
    print("The animal is classified as a dog.")  
else:  
    print("The animal is classified as a cat.")
```

The animal is classified as a cat.

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